Mukul Sangwan

LinkedIn GitHub
Email: sangwanmukul123@gmail.com
Mobile: +91 7419822000

Education

- **Vellore Institute of Technology, Chennai** Chennai, India
  Bachelor of Technology in Computer Science and Engineering; ***CGPA: 9.04*** *Aug. 2022 - Jul. 2026*
- **RPS Sr. Sec. School,Khatod** Mohindergarh, India
  CBSE; ***Class XIIth: 94.2%, Class Xth: 95.8%*** *Apr. 2018 – Jul. 2022*

Experience

Decimal Technologies Pvt. Ltd. Gurugram, Haryana
Development Intern *May 2026 – Present*

- Developing enterprise-grade backend solutions for **Federal Bank** using **Java (Spring Boot)** and **Python**, supporting automated loan processing, document validation, and business rule execution.
- Designed and implemented configurable **FTR Gatekeeping Validator** workflows, REST APIs, and rule registry modules using **Java 17, Spring Boot, MySQL, Maven, and JPA/Hibernate**.
- Built and enhanced **Python**-based utilities for data processing, validation, automation, and analysis to streamline internal workflows and improve operational efficiency.
- Developed scheduler-based parallel loan processing, optimized SQL queries, resolved production issues, and collaborated using **Git**, **Linux**, **DBeaver**, and **Antigravity IDE** in an Agile development environment.

Research & Publication

L-MACD: Learning-Based Multi-Modal Adaptive CAPTCHA Defense Against Multimodal AI Attacks
IEEE Access *(2026)* ***View Paper***

- Published in **IEEE Access**, developed an adaptive CAPTCHA system reducing bot success rate to **8%** while maintaining **93% human usability**.
- Designed a **multi-agent framework** and trained **Random Forest models** to optimize CAPTCHA parameters using the AGE metric (SR_{human} - SR_{bot}).
- Integrated **statistical validation (t-test, ANOVA)** and **SHAP-based explainability** to analyze model performance and feature importance.

Projects

Multimodal Ozone Layer Depletion Prediction System
Python, Scikit-learn, PyTorch Geometric, XGBoost

- Built machine learning models using **Random Forest, SVM, and XGBoost** along with **Graph Neural Networks (GCN, GraphSAGE)** to model spatial relationships in environmental data.
- Developed a pipeline including **data preprocessing, feature engineering, and graph construction** for ozone prediction using multimodal datasets.

Generative Data Augmentation with MARL
Techstack : *PyTorch, TensorFlow, GANs, RLlib*

- Designed a **reinforcement learning-based framework** for data augmentation in low-sample scenarios.
- Integrated **GANs with RL techniques** to generate synthetic data, improving model generalization compared to traditional sampling methods.

Achievements & Certifications

- **Coursera Project Network:**Certified in **Introduction to Android Mobile Application Development** by Meta, issued in June 2023.
- **Microsoft Azure:**Certified **DP-900: Microsoft Azure Data Fundamentals**, July 2024.
- **Coursera Project Network:**Certified in **SQL for Data Science** by University of California, Davis, issued in Sep. 2024.
- **Udemy:**Completed **Full-Stack Web Development Bootcamp**, Apr. 2025.

Technical Skills

- **Programming Languages:** C++, Python, Java, SQL, JavaScript
- **Data Structures & Algorithms:** Arrays, Linked Lists, Trees, Graphs, Dynamic Programming, Greedy, Recursion
- **Core CS:** Operating Systems, Computer Networks, DBMS, OOPS
- **Machine Learning & AI:** PyTorch, TensorFlow, Scikit-learn, Reinforcement Learning, SHAP